SmartVision: A Deep Learning and IoT-Based Framework for Automated Kitchen Hygiene Monitoring

CAPSTONE PROJECT REPORT

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DECLARATION

We affirm that the project work titled **"SmartVision: A Deep Learning and IoT-Based Framework for Automated Kitchen Hygiene Monitoring"** being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

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BONAFIDE CERTIFICATE

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We are deeply grateful to the late **"Kalvivallal" Thiru T. Kalasalingam,** Chairman of the Kalasalingam Group of Institutions, and to late "**Illayavallal" Dr. K. Sridharan**, Chancellor, as well as **Dr. S. Shasi Anand**, Vice President, who has been a guiding light in all our university's endeavours.

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PROJECT SUMMARY

Project Title	**SmartVision: A Deep Learning and IoT-Based Framework for Automated Kitchen Hygiene Monitoring**	
Project Team Members (Name with Register No)	**Rajesh Kannan L - 99220041561** **Surarapu Ashish - 99220041579** **Jeya Mithra K - 99220041618** **Meenakshi B - 99220041666**	
Guide Name/Designation	**Dr.T.Dhiliphan Rajkumar, Associate Professor, Department of Computer Science and Engineering, KARE**	
Program Concentration Area	Computer Vision, Web Application Development, Real-Time System, Database Management	
Technical Requirements	Python, OpenCV, YOLO (Ultralytics), Flask, SQLite, NumPy, HTML, CSS, SMTP (Email Service)	
Engineering standards and realistic constraints in these areas		
Area	**Codes & Standards / Realistic Constraints**	**Tick ✓**
Economic	System development involves costs related to hardware setup (camera), model deployment, maintenance, and hosting of the web application.	✓
Environmental	Continuous operation of camera and computing systems leads to energy consumption, especially during real-time video processing and model inference.	✓
Social	User acceptance depends on trust in AI-based monitoring systems and ease of use of the web interface.	✓
Ethical	The system must ensure data privacy, secure storage of user information, and responsible use of surveillance without misuse.	✓

Health and Safety	Incorrect detection or missed alerts may lead to safety risks; therefore, system accuracy and timely alerts are critical.	✓
Manufacturability	Requires integration of software components such as deep learning models, web framework, and database into a unified system.	✓
Sustainability	The system requires periodic model updates, maintenance, and system monitoring to ensure long-term reliability and performance.	✓

REALISTIC CONSTRAINTS:

Environmental:

The SmartVision system relies on continuous operation of camera modules, computing devices, and network infrastructure for real-time monitoring. The use of video processing, deep learning inference, and web-based services contributes to energy consumption. Although the system reduces manual inspection efforts and improves operational efficiency, it still depends on electronic hardware and continuous power supply. The environmental impact is therefore associated with computational resources, camera usage, and network dependency required for real-time detection.

Sustainability:

Sustainability is an important factor as the SmartVision system requires regular maintenance, including updating deep learning models and managing system performance. YOLO-based models must be retrained or fine-tuned periodically to adapt to new environments and improve detection accuracy. Additionally, long-term system usage depends on reliable hardware such as cameras and servers. Without proper updates and maintenance, the detection accuracy and system efficiency may degrade over time.

Engineering Standards:

Software Development Standards (IEEE 829, IEEE 830, IEEE 12207): The system follows structured development practices including requirement specification, testing, and deployment. The Flask-based application ensures modular backend development, while the front-end dashboard supports user interaction and monitoring.

Computer Vision and AI Standards: The system utilizes YOLO-based deep learning models for real-time object detection. These models are designed to provide high accuracy and low latency, making them suitable for continuous monitoring environments.

Web and Security Standards: User authentication is implemented using secure password hashing techniques, and data is stored in a structured SQLite database. The system ensures secure handling of user credentials and detection logs.

Communication and Alert Standards: The system uses SMTP protocol for sending automated email alerts, ensuring timely notification of detected events.

Documentation Standards (IEEE 1063): Project documentation is maintained with clarity and consistency to ensure easy understanding and evaluation.

SUSTAINABLE DEVELOPMENT GOALS

SDG 3: Good Health and Well-being – Ensures healthy lives and promotes well-being for all.

SDG 6: Clean Water and Sanitation – Ensures safe and sustainable water and sanitation for all.

SDG 9: Industry, Innovation and Infrastructure - Advances AI applications and supports digital creativity infrastructure.

SDG 12: Responsible Consumption and Production – Promotes efficient use of resources and reduces waste.

ABSTRACT

The demand for automated monitoring systems has grown significantly due to increasing requirements for safety and hygiene in real-time environments. Traditional monitoring methods rely heavily on manual supervision, which is often inefficient, inconsistent, and prone to human error, making them unsuitable for continuous and large-scale monitoring tasks.

To address these limitations, this project proposes SmartVision, an intelligent real-time monitoring system that integrates deep learning-based computer vision with a web-based alert and management platform. The system uses a camera module to capture live video streams, which are processed using YOLO-based models implemented with OpenCV and the Ultralytics framework, enabling accurate object detection with low latency.

Detected events are continuously analyzed, and alerts are triggered when predefined conditions persist beyond a specified time threshold. The system is integrated with a Flask-based web application that provides user authentication, real-time video streaming, and an interactive dashboard for monitoring detection results. Additionally, detected events are stored in a database, and an automated email alert system using SMTP is implemented to notify users during critical situations.

The performance of the system is evaluated using metrics such as accuracy, precision, recall, F1-score, detection latency, false positive rate, system throughput, and alert response time. By combining deep learning, real-time video processing, and web technologies, SmartVision provides a scalable, efficient, and user-friendly solution that reduces dependency on manual supervision while ensuring timely detection and response in safety-critical environments.

KEYWORDS: Deep Learning, YOLO, Computer Vision, Real-Time Monitoring, Flask, Email Alerts, Smart Surveillance.

TABLE OF CONTENTS

TITLE		PAGE NO.
ABSTRACT		**viii**
LIST OF FIGURES		**ix**
LIST OF TABLES		**x**
LIST OF ACADEMIC REFERENCE COURSES		**xi**
CHAPTER I	**INTRODUCTION**	**1**
1.1	Problem Statement	**2**
1.2	Objectives of the Project	**4**
1.3	Scope of the Project	**5**
1.4	Methodology Overview	**6**
1.5	Organization of the Report	**8**
CHAPTER II	**LITERATURE REVIEW**	
2.1	Overview of Related Work	**9**
2.2	Review of Similar Projects or Research Papers	**11**
2.3	Summary and Gap Identification	**12**
CHAPTER III	**SYSTEM ANALYSIS**	
3.1	Requirements Gathering	**15**
3.2	System Behavior Requirements	**16**
3.4	Feasibility Study 3.4.1 Technical Feasibility 3.4.2 Operational Feasibility 3.4.3 Economic Feasibility	**17**
3.5	Risk Analysis	**19**
CHAPTER IV	**SYSTEM DESIGN**	
4.1	SmartVision System Architecture	**20**
4.2	Module Design	**23**
4.3	Database Design	**24**
4.3.1	Database Schema	**25**
4.4	User Interface Design	**27**

CHAPTER V	**IMPLEMENTATION**	
5.1	Technology Stack	**29**
5.1.1	Programming Languages and Tools	**30**
5.2	Implementation of Modules	**31**
5.2.1	Data Acquisition Model	**31**
5.2.2	Vision Based Detection Module	**31**
5.2.3	Sensor Processing Module	**32**
5.2.4	Data Fusion and Decision Module	**32**
5.2.5	Alert and Notification Module	**32**
5.2.6	Storage and Data Management Module	**33**
5.3	Integration of Modules	**33**
CHAPTER VI	**TESTING**	
6.1	Testing Methodology	**35**
6.2	Test Cases and Results	**36**
6.3	Bug Tracking and Resolution	**37**
CHAPTER VII	**RESULTS AND DISCUSSION**	
7.1	System Output Screenshots	**38**
7.2	Evaluation Metrics	**41**
7.3	Comparison with Existing Systems	**42**
7.4	Challenges Faced	**43**
7.5	Solutions and Improvements	**44**
CHAPTER VIII	**CONCLUSION & FUTURE SCOPE**	
8.1	Conclusion	**44**
8.2	Future Scope	**45**
REFERENCES : List of books, articles, research papers, and other resources used		**46**
PUBLICATION		

CERTIFICATIONS
PLAGIARISM REPORT

LIST OF FIGURES

FIGURES	DETAILS	PAGE NO.
Figure 4.3	Architectural diagram	33
Figure 7.1	Live Camera Detection	37
Figure 7.2	Email Notification	37
Figure 7.3	Dash board of Smart Vision	38
Figure 7.4	Live Camera Detection	39
Figure 7.5	Detection Alerts	40

LIST OF TABLES

TABLES	DETAILS	PAGE NO.
7.1	Summary of System Performance Metrics	44
7.2	Model Metrics	44

LIST OF ACADEMIC REFERENCE COURSES

S. NO.	COURSE CODE	COURSE NAME
1.	213CSE1301	Introduction to Artificial Intelligence and Machine learning
2.	213CSE3301	Deep Learning
3.	212ECE1401	IOT sensor and devices

CHAPTER – I
INTRODUCTION

Kitchen hygiene plays a critical role in ensuring food safety, maintaining public health standards, and preventing foodborne illnesses. Contamination arising from unclean surfaces, improper waste management, and unsafe handling practices can lead to serious health hazards, especially in environments such as restaurants, hostels, and institutional kitchens where food is prepared at scale. According to global food safety studies, a significant percentage of foodborne diseases originate from poor hygiene practices during food preparation, highlighting the need for continuous and reliable monitoring systems. Traditionally, kitchen hygiene is maintained through manual inspection and supervision. Staff members or inspectors visually examine kitchen areas such as countertops, cooking zones, utensils, and waste bins to assess cleanliness and safety. While this approach is simple and widely adopted, it suffers from several inherent limitations. Manual inspections are time-consuming, subjective, and inconsistent, often depending on the attentiveness and experience of the individual performing the inspection. In high-activity environments, such as commercial kitchens during peak hours, it becomes increasingly difficult to monitor all areas continuously, leading to overlooked hygiene violations such as spills, clutter, or waste overflow.

In recent years, technological advancements in Artificial Intelligence (AI) and the Internet of Things (IoT) have opened new possibilities for automating monitoring and decision-making processes. Deep learning, a subset of AI, has shown remarkable success in computer vision tasks such as object detection and scene understanding. Models based on the YOLO (You Only Look Once) architecture are particularly well-suited for real-time applications due to their ability to process images quickly while maintaining high detection accuracy. These models can be trained to identify hygiene-related issues such as unclean surfaces, clutter, and unsafe conditions directly from visual data.

At the same time, IoT technologies enable real-time monitoring of environmental and operational parameters using interconnected sensors. For example, ultrasonic sensors can measure waste-bin levels, while power sensors can track appliance usage patterns. These sensors provide valuable contextual information that cannot be captured through visual analysis alone. However, systems that rely solely on sensors are limited in their ability to detect surface-

level hygiene issues, while vision-only systems lack awareness of underlying operational conditions.

This gap between vision-based and sensor-based approaches highlights the need for an integrated and intelligent monitoring framework that combines the strengths of both technologies. By fusing visual intelligence with sensor data, it is possible to achieve a more comprehensive understanding of kitchen hygiene conditions, reduce false detections, and improve decision reliability. The motivation for developing the SmartVision system arises from the need to overcome the shortcomings of existing approaches and to provide a low-cost, scalable, and automated solution for real-time kitchen hygiene monitoring. The proposed system aims to minimize human intervention, ensure continuous surveillance, and enable timely detection of hygiene violations. Such a system is particularly valuable in resource-constrained environments where maintaining consistent hygiene standards is challenging.

Furthermore, the increasing adoption of smart technologies in everyday environments, including smart homes and smart kitchens, creates an opportunity to integrate hygiene monitoring as a core component of intelligent infrastructure. By leveraging edge computing and lightweight deep learning models, the system can operate efficiently in real-time without heavy reliance on cloud resources, making it practical for widespread deployment. In summary, the motivation behind this work is to address the critical need for reliable, continuous, and automated kitchen hygiene monitoring by combining advancements in AI and IoT, thereby enhancing food safety, reducing human error, and promoting healthier living environments.

1.1 Problem Statement

Maintaining consistent hygiene standards in kitchen environments is a critical yet challenging task, particularly in domestic kitchens, hostels, and small to medium-scale commercial food preparation units. Despite its importance in ensuring food safety and preventing contamination, kitchen hygiene monitoring is still predominantly carried out through manual inspection methods. These traditional approaches suffer from multiple limitations that reduce their effectiveness and reliability.

One of the primary issues with manual monitoring is its dependency on human observation, which introduces subjectivity and inconsistency. The quality of inspection often varies based on the experience, attentiveness, and workload of the individual responsible. During peak operational hours, it becomes difficult for staff to continuously monitor all areas

of the kitchen, resulting in overlooked hygiene violations such as spills, cluttered workspaces, unclean surfaces, and overflowing waste bins. These unnoticed issues can accumulate over time and significantly increase the risk of contamination and foodborne illnesses.

Another major limitation is the lack of continuous real-time monitoring. Traditional inspection methods are typically periodic rather than continuous, meaning hygiene issues that arise between inspections may remain undetected for extended periods. This delay in detection can lead to unsafe kitchen conditions and delayed corrective actions, ultimately affecting food quality and safety.

Existing technological solutions also fail to fully address this problem. Systems that rely solely on IoT sensors are capable of monitoring specific parameters such as waste levels or appliance usage but lack the ability to detect visual hygiene issues like spills, clutter, or improperly cleaned surfaces. On the other hand, vision-based systems using computer vision can detect visible anomalies but lack contextual awareness of operational conditions, such as whether a waste bin is nearing overflow or if an appliance is being used abnormally. As a result, both approaches, when used independently, provide only a partial view of the overall hygiene condition.

Furthermore, many existing solutions are fragmented and domain-specific, focusing on isolated aspects such as hand hygiene, PPE compliance, or waste monitoring rather than providing a holistic assessment of kitchen hygiene. This lack of integration limits their effectiveness in real-world environments where multiple factors contribute to hygiene and safety. Therefore, the core problem addressed in this project is the absence of a unified, real-time, automated, and cost-effective kitchen hygiene monitoring system that can:

- Continuously monitor both visual and operational hygiene conditions
- Detect a wide range of hygiene violations accurately
- Reduce dependency on manual inspection
- Provide timely alerts for immediate corrective action
- Operate efficiently in real-world, resource-constrained environments

To overcome these challenges, there is a need for an integrated framework that combines deep learning–based computer vision with IoT sensor technologies, enabling comprehensive, reliable, and real-time kitchen hygiene monitoring.

1.2 Objectives of the Project

The primary objective of this project is to design and implement SmartVision, an intelligent, automated, and real-time kitchen hygiene monitoring system that integrates deep learning–based computer vision with IoT-enabled sensing. The system is intended to address the limitations of traditional manual inspection methods by providing continuous, accurate, and context-aware monitoring of hygiene conditions in kitchen environments.

The first objective is to develop a fully automated system capable of continuously monitoring kitchen environments without human intervention. Unlike traditional inspection methods that operate periodically, the proposed system is designed to function in real time, ensuring that hygiene violations are detected immediately as they occur. This continuous monitoring approach helps in minimizing delays in detection and enables proactive maintenance of hygiene standards.

A key objective is to design and implement a robust computer vision module using deep learning techniques. The system employs a YOLO-based object detection model to analyze real-time video streams captured from kitchen environments. This module is trained to detect various hygiene-related conditions, including spills, cluttered surfaces, unclean areas, and uncovered food. The objective is not only to achieve high detection accuracy but also to ensure low-latency inference so that the system can operate efficiently in dynamic and fast-paced kitchen settings. The use of deep learning enables the system to adapt to variations in lighting, background complexity, and human activity.

Another important objective is to incorporate IoT sensors to monitor operational aspects of kitchen hygiene that cannot be detected visually. The system integrates an ultrasonic sensor to measure waste-bin levels and identify overflow conditions, and a power sensor to monitor appliance usage patterns. The objective is to provide contextual information about kitchen operations, such as delayed waste disposal or abnormal appliance usage, which may indicate potential hygiene or safety risks. The sensor data is collected and transmitted using a NodeMCU (ESP8266) microcontroller, ensuring efficient and real-time communication.

The project aims to develop a hybrid framework that combines outputs from the vision module and IoT sensors through a data fusion mechanism. Instead of treating these data sources independently, the system correlates visual and sensor information to improve decision accuracy.

The objective is to implement an intelligent decision engine that applies rule-based logic and threshold analysis to evaluate combined inputs. This approach reduces false positives, enhances reliability, and provides a more comprehensive understanding of kitchen hygiene conditions. An essential objective is to design an effective alert mechanism that notifies users immediately when a hygiene violation or abnormal condition is detected. The system provides feedback through an LCD display that shows status messages and a buzzer that generates audible alerts.

The goal is to ensure that alerts are clear, timely, and easily noticeable, enabling users to take immediate corrective actions. This real-time feedback system plays a crucial role in maintaining continuous hygiene compliance. The project emphasizes the development of a low-cost system using affordable and readily available hardware components. The objective is to make the solution accessible for a wide range of users, including households, hostels, and small-scale commercial kitchens. The objective also includes supporting future enhancements such as cloud integration, predictive analytics, and centralized monitoring across multiple kitchen environments. The final objective is to minimize the dependency on manual supervision and reduce human errors associated with traditional inspection methods. By providing consistent, objective, and automated monitoring, the system aims to improve overall hygiene standards, enhance food safety, and promote healthier kitchen environments.

1.3 Scope of the Project

The scope of the SmartVision project encompasses the design, development, and deployment of an intelligent and automated system for monitoring hygiene conditions in kitchen environments using a combination of deep learning and Internet of Things (IoT) technologies. The system is intended to operate in real time, providing continuous observation and analysis of both visual and operational aspects of kitchen hygiene. It is designed to be applicable across a wide range of environments, including domestic kitchens, hostels, small-scale restaurants, and institutional food preparation areas, where maintaining consistent hygiene standards is essential but often challenging due to reliance on manual supervision.

The project primarily focuses on detecting hygiene-related issues that are commonly observed in kitchen settings. Through the integration of a computer vision module, the system analyzes live video streams to identify conditions such as spills, cluttered workspaces, unclean surfaces, and improperly handled or uncovered food. At the same time, the system incorporates IoT-based sensors to monitor operational parameters, including waste-bin levels and appliance

usage patterns. This dual approach enables the system to capture both visible and non-visible hygiene factors, thereby providing a more comprehensive understanding of the kitchen environment.

The system operates on an edge-based architecture, ensuring that data processing and decision-making occur locally with minimal latency. This allows for immediate detection of hygiene violations and prompt alert generation through visual and audio feedback mechanisms. The scope also includes the storage of detected events and sensor data, which can be used for further analysis, performance evaluation, and potential future enhancements such as trend monitoring and predictive analytics.

From a technical perspective, the project covers the implementation of a YOLO-based deep learning model for real-time object detection, along with the integration of hardware components such as a NodeMCU microcontroller, ultrasonic sensor, and power sensor. The system is designed with a modular structure, allowing for scalability and future expansion, including the addition of new sensors, extended detection capabilities, and integration with cloud-based platforms or user interfaces.

However, the scope of the project is limited to monitoring and alerting functionalities. The system is not designed to take autonomous corrective actions or control kitchen equipment. Its effectiveness is influenced by factors such as camera placement, lighting conditions, and the diversity of the training dataset. Additionally, advanced features such as mobile applications, cloud-based analytics, and detection of more complex hygiene parameters like pest presence or food spoilage are considered beyond the current scope and are identified as areas for future work.

1.4 Methodology Overview

The SmartVision system follows a structured and integrated methodology that combines deep learning–based computer vision with IoT-enabled sensing to achieve real-time kitchen hygiene monitoring. The methodology is designed as a continuous and automated pipeline, where data is acquired, processed, analyzed, and translated into actionable alerts. This approach ensures that both visual and operational aspects of kitchen hygiene are monitored simultaneously, providing a comprehensive and reliable assessment.

The process begins with data acquisition, where the system collects inputs from two primary sources: a camera and IoT sensors. The camera captures continuous video streams of key kitchen areas such as countertops, cooking zones, and waste bins. At the same time, IoT components connected to a NodeMCU microcontroller gather sensor data, including waste-bin levels through an ultrasonic sensor and appliance usage through a power sensor. These inputs form the foundation for subsequent analysis.

Once the data is collected, it undergoes preprocessing to ensure consistency and accuracy. Visual data is resized and normalized to match the input requirements of the deep learning model, while data augmentation techniques are applied during training to improve robustness against variations in lighting and environmental conditions. Sensor data is filtered and normalized to eliminate noise and ensure stable readings. Additionally, time synchronization mechanisms are applied to align visual frames with corresponding sensor readings, enabling accurate correlation in later stages.

The core of the methodology lies in the vision-based hygiene detection module, which utilizes a YOLO-based deep learning model to analyze real-time images. The model identifies hygiene-related conditions such as spills, cluttered surfaces, unclean areas, and uncovered food by generating bounding boxes, class labels, and confidence scores. Its single-stage architecture allows for fast and efficient inference, making it suitable for real-time deployment in dynamic kitchen environments. Parallel to this, the IoT sensor processing module continuously evaluates sensor data to detect operational anomalies. The ultrasonic sensor measures waste-bin levels and determines whether the bin is approaching overflow, while the power sensor tracks appliance usage patterns to identify abnormal or prolonged operation. These sensor-based observations provide contextual insights that complement the visual analysis.

A key component of the methodology is the data fusion and decision-making layer, where outputs from the vision module and sensor module are integrated. Instead of treating these inputs independently, the system correlates them to improve decision accuracy and reduce false positives. For example, a hygiene alert related to waste overflow is validated using both sensor thresholds and visual confirmation. This fusion-based approach ensures that alerts are generated based on multiple factors, enhancing reliability and contextual understanding.

Following this, the system performs alert generation and user feedback, where detected violations trigger immediate notifications. Alerts are displayed through an LCD screen,

providing clear and concise information about the detected issue, while a buzzer generates an audible signal to capture user attention. This real-time feedback mechanism enables prompt corrective action without requiring continuous manual monitoring.

1.5 Organization of the Report

This report is organized into several chapters that provide a detailed explanation of the development and implementation of the Kalaimayam system.

Chapter 1 introduces the project by discussing the background, motivation, problem statement, objectives, scope, and methodology of the system. Chapter 2 presents the literature survey, which reviews existing research and technologies related to AI-based image generation and neural style transfer.

Chapter 3 describes the system analysis and explains the requirements of the proposed system. Chapter 4 focuses on the system design, including the architecture, module structure, and user interface design.

Chapter 5 explains the implementation of the system and discusses the technologies and tools used during development. Chapter 6 presents the results, system evaluation, and comparison with existing systems. Finally, Chapter 7 concludes the report by summarizing the project findings and discussing potential future enhancements.

CHAPTER-II

LITERATURE REVIEW

2.1 Overview of Related Work

The rapid advancement of Artificial Intelligence (AI), computer vision, and Internet of Things (IoT) technologies has significantly influenced the development of automated monitoring systems in domains such as healthcare, industrial safety, and food hygiene. In the context of kitchen environments, several research efforts have been made to improve hygiene monitoring and safety compliance using both traditional and modern technological approaches. This section provides an overview of existing work relevant to automated kitchen hygiene monitoring, highlighting key contributions and identifying existing gaps.

Early research in kitchen monitoring primarily relied on traditional image processing techniques, such as edge detection, background subtraction, and Haar Cascade classifiers, to identify objects and human activities. These methods demonstrated initial success in controlled environments but suffered from major limitations when applied to real-world kitchen settings. Variations in lighting conditions, occlusions caused by human movement, and complex backgrounds often led to reduced accuracy and unreliable performance. As a result, these approaches were not suitable for continuous and robust hygiene monitoring.

The emergence of deep learning-based computer vision marked a significant improvement in visual monitoring systems. Convolutional Neural Networks (CNNs) enabled more accurate feature extraction and object recognition, paving the way for advanced detection models. Among these, the YOLO (You Only Look Once) family of models became widely popular due to their ability to perform real-time object detection with high accuracy and low latency. Recent studies have explored different YOLO variants, such as YOLOv5, YOLOv8, and YOLOv10, demonstrating improvements in precision, robustness, and inference speed. These models have been successfully applied to detect hygiene-related conditions, including spills, clutter, and unsafe practices in kitchen environments.

Parallel to vision-based approaches, IoT-based monitoring systems have gained attention for tracking environmental and operational parameters. Sensor-based systems have been developed to monitor waste-bin levels, temperature, humidity, and appliance usage. These systems are capable of generating real-time alerts and improving operational efficiency.

However, IoT-only systems lack the ability to detect visual hygiene issues such as spills, cluttered surfaces, or improper food handling, which are critical for maintaining cleanliness.

More recently, some studies have attempted to combine AI and IoT technologies for smarter monitoring solutions. These hybrid systems aim to integrate sensor data with intelligent analysis to provide better decision-making. Despite this progress, most existing solutions remain fragmented and narrowly focused, addressing only specific aspects of kitchen hygiene rather than providing a comprehensive, unified system. Additionally, many systems are evaluated in controlled environments, limiting their applicability in real-world kitchens characterized by dynamic conditions, continuous movement, and varying lighting.

A critical analysis of the existing literature reveals that there is still a lack of integrated, real-time, and scalable frameworks that can simultaneously monitor both visual and operational hygiene factors. This gap highlights the need for a system that combines the strengths of deep learning–based computer vision and IoT sensor technologies to achieve a more accurate and reliable assessment of kitchen hygiene.

The proposed Smart Vision system is motivated by these limitations and aims to address them by providing a unified framework that integrates real-time visual detection with sensor-based monitoring. By leveraging recent advancements in deep learning and IoT, the system seeks to overcome the shortcomings of existing approaches and deliver a comprehensive solution for automated kitchen hygiene monitoring.

In recent years, artificial intelligence has significantly influenced the field of digital image processing and creative media generation. Many research studies have explored how deep learning models can be used to generate artistic images automatically. Techniques such as Neural Style Transfer, Generative Adversarial Networks (GANs), and diffusion models have made it possible for computers to create visually appealing images by learning patterns from large image data sets.

Several AI-based platforms have been developed to assist artists, designers, and content creators in generating digital artwork. These systems allow users to transform ordinary images into artistic paintings by applying the style of famous artworks. Other systems allow users to generate completely new images from text descriptions using advanced generative models.

2.2 Review of Similar Projects or Research Papers

YOLOv10 has been explored as an advanced detection framework for improving performance in complex environments. Liu et al. [1] proposed an improved YOLOv10-based model specifically designed for kitchen hygiene and safety monitoring. Their approach focuses on addressing real-world challenges such as occlusion and lighting variations, thereby enhancing detection accuracy and robustness in real-time scenarios.

Similarly, Radhakrishnan et al. [2] developed an automated alert system using machine vision techniques to monitor hygiene conditions in restaurant environments. While their system successfully detects unsafe conditions and generates alerts, it lacks deep integration with advanced detection pipelines and scalability, limiting its applicability in large-scale systems.

Comparative studies have also been conducted to evaluate the performance of different YOLO models. Geetha and Hussain [3] analyzed YOLOv5, YOLOv8, and YOLOv10, highlighting improvements in speed, detection accuracy, and computational efficiency. Their findings confirm that newer YOLO versions provide better performance for real-time applications.

In the domain of safety compliance, Prabha et al. [4] developed a computer vision-based system for detecting personal protective equipment (PPE) such as gloves, masks, and helmets. Although effective in ensuring workplace safety, the system is limited to specific object categories and does not support broader monitoring capabilities. Similarly, Kumar et al. [5] implemented a face mask detection system using machine learning techniques, which performs well for mask compliance but focuses on a single safety parameter.

Chen et al. [6] introduced a deep learning-based pest detection system for smart kitchen environments. While their approach contributes to hygiene monitoring, it lacks integration with multi-functional monitoring systems. Zhou et al. [7] further explored lightweight detection models by developing a YOLOv5s-based embedded system for detecting kitchen dress compliance. Although optimized for edge devices, the system is limited in scope and does not support comprehensive monitoring.

Shi et al. [8] proposed a real-time mask detection system using YOLOv5 combined with edge computing, focusing on low-latency processing. While this improves speed, the system remains restricted to a single detection task. In contrast, traditional methods such as

Haar Cascade classifiers, used by Bernabe et al. [9], offer computational efficiency but suffer from low accuracy and poor adaptability in dynamic environments.

Earlier research has also explored deep learning approaches for food safety monitoring. Tao et al. [10] developed a system capable of detecting hygiene violations in kitchen environments, demonstrating strong detection capabilities. However, the absence of real-time alert mechanisms limits its effectiveness. Similarly, van Staden and Brown [11] evaluated YOLO-based algorithms for hand detection, but their work focuses on a narrow use case.

Protik et al. [12] proposed a YOLOv4-based PPE detection system using TensorFlow, achieving reliable real-time performance. However, like many existing systems, it lacks integration with broader monitoring frameworks. Zhang et al. [13] implemented Faster R-CNN for PPE detection, achieving high accuracy but at the cost of slower processing speed compared to YOLO-based models.

Domain-specific safety systems have also been developed. Delhi et al. [14] created a PPE compliance detection system for construction sites, which ensures worker safety but lacks adaptability to other environments. Additionally, Galkin et al. [15] proposed a vision-based hygiene monitoring system focused on handwashing procedures. While effective for specific tasks, it lacks scalability and multi-object detection capabilities.

2.3 Summary and Gap Identification

The review of existing literature highlights the significant progress made in applying deep learning, computer vision, and IoT technologies for kitchen monitoring and safety applications. Researchers such as Athulya Sundaresan Geetha and Muhammad Hussain (2024) demonstrated the effectiveness of advanced YOLO models in achieving high accuracy and real-time performance for kitchen safety detection. Similarly, Z. Zhou et al. (2023) and I. C. Chen et al. (2024) focused on monitoring hygiene compliance among kitchen staff, showing that deep learning models can reliably detect the use of protective equipment and adherence to hygiene practices. In addition, works by Norisza Dalila Ismail et al. (2023) emphasized the importance of safety-focused detection systems, such as fire detection, while Iker Zhong et al. (2022) explored human activity monitoring using object detection and tracking techniques.

Parallel to these vision-based approaches, IoT-based systems have been developed to monitor environmental and operational parameters such as waste levels, gas leakage, temperature, and appliance usage. These systems provide real-time alerts and improve safety

and efficiency in kitchen environments. However, they are inherently limited in their ability to capture visual hygiene conditions, which are critical for assessing cleanliness.

Although these studies demonstrate the potential of AI and IoT technologies, a careful analysis reveals several important gaps. Most existing systems are narrow in scope, focusing on isolated aspects such as personnel compliance, safety hazards, or environmental monitoring. There is a lack of a holistic approach that considers kitchen hygiene as a multi-dimensional problem involving both visual and operational factors.

Furthermore, vision-only systems are limited by their inability to capture contextual information such as waste accumulation or appliance usage, while sensor-only systems fail to detect visible hygiene violations like spills, clutter, or unclean surfaces. This separation leads to incomplete analysis and reduces the reliability of decision-making.

Another critical gap lies in the lack of integration and data fusion mechanisms. Most existing solutions treat visual and sensor data independently rather than combining them to enhance accuracy and reduce false positives. Additionally, many systems are evaluated under controlled laboratory conditions and may not perform effectively in real-world kitchen environments characterized by dynamic activity, lighting variations, and occlusions. There are also challenges related to scalability and cost-effectiveness, as some solutions require complex infrastructure or high computational resources, making them unsuitable for small-scale kitchens or resource-constrained environments.

To address these limitations, there is a clear need for a unified, real-time, and scalable system that integrates deep learning–based visual detection with IoT sensor data. The proposed SmartVision system is designed to fill this gap by combining both approaches into a single framework, enabling comprehensive monitoring of kitchen hygiene. By incorporating data fusion, real-time alerting, and edge-based processing, the system aims to provide a reliable, cost-effective, and practical solution for improving hygiene standards in diverse kitchen environments.

CHAPTER-III

SYSTEM ANALYSIS

3.1 Requirements Gathering

Requirements gathering is a crucial phase in the development of the SmartVision system, as it defines the functional and non-functional needs of the proposed solution. This phase involves identifying the expectations of users, understanding real-world challenges in kitchen environments, and determining the technical specifications required to design an effective and reliable hygiene monitoring system.

The requirements for this project were identified through an analysis of common issues observed in domestic and commercial kitchens, such as inconsistent hygiene practices, delayed waste disposal, and lack of continuous monitoring. In addition, insights from existing research and technological advancements in deep learning and IoT were considered to ensure that the system addresses current limitations and provides a practical solution.

From a functional perspective, the system is required to continuously monitor kitchen environments in real time. It should be capable of capturing live video input through a camera and analyzing it using a deep learning model to detect hygiene-related issues such as spills, cluttered surfaces, unclean areas, and uncovered food. Simultaneously, the system must collect data from IoT sensors to monitor operational parameters, including waste-bin levels and appliance usage. The integration of these components should enable the system to generate accurate and context-aware decisions. Furthermore, the system must provide immediate alerts to users through visual and audio feedback mechanisms, ensuring timely corrective actions.

Scalability and flexibility are also key considerations. The system should be designed in a modular manner, allowing additional sensors or advanced features to be integrated in the future without significant redesign. Ease of use is another important factor, as the system should be simple to install, operate, and maintain, even for users with limited technical expertise.

Hardware requirements for the system include a camera for capturing visual data, a NodeMCU (ESP8266) microcontroller for processing and communication, an ultrasonic sensor

for waste monitoring, a power sensor for appliance tracking, an LCD display for user feedback, and a buzzer for alert generation. On the software side, the system requires a deep learning framework for implementing the YOLO-based detection model, along with programming tools such as Python for model development and integration.

Overall, the requirements gathering phase ensures that the SmartVision system is designed to meet both user needs and technical constraints. By clearly defining these requirements, the project establishes a strong foundation for developing a reliable, efficient, and scalable kitchen hygiene monitoring solution.

3.2 System Behavior Requirements

Functional requirements define the core operations and capabilities that the SmartVision system must perform to achieve its objective of automated kitchen hygiene monitoring. These requirements describe how the system interacts with inputs, processes data, and produces outputs to ensure effective and reliable performance in real-time kitchen environments.

The primary functional requirement of the system is to enable continuous real-time monitoring of kitchen environments. The system must be capable of capturing live video streams through a camera installed in key areas such as countertops, cooking zones, and waste disposal regions. These video inputs serve as the foundation for visual analysis and must be processed continuously without interruption.

Another critical requirement is the implementation of a vision-based detection mechanism using a deep learning model. The system must analyze incoming video frames and accurately detect hygiene-related conditions such as spills, cluttered surfaces, unclean areas, and uncovered food. It should generate outputs in the form of bounding boxes, labels, and confidence scores, allowing precise identification of hygiene violations. The detection process must be efficient enough to support real-time operation.

In parallel, the system must perform IoT-based sensor monitoring to capture operational data from the kitchen environment. The ultrasonic sensor is required to measure waste-bin levels and determine whether the bin is approaching overflow. Similarly, the power sensor must monitor appliance usage and identify abnormal patterns, such as excessive or prolonged

operation. These sensor readings must be collected periodically and transmitted to the processing unit for further analysis.

A key functional requirement is the ability to perform data integration and fusion. The system must combine outputs from the vision module and sensor module to generate more accurate and context-aware decisions. For instance, a waste overflow alert should ideally be triggered based on both sensor thresholds and visual confirmation. This integration helps in reducing false detections and improving overall system reliability.

The system must also include a decision-making mechanism that evaluates the processed data and determines whether a hygiene violation or abnormal condition exists. This mechanism should apply predefined rules, thresholds, and logic to classify the severity of detected issues and decide when to trigger alerts.

Another essential requirement is the implementation of a real-time alert generation system. When a hygiene violation is detected, the system must immediately notify users through an LCD display, providing clear and concise information about the issue. Additionally, a buzzer must be activated to provide an audible alert, ensuring that users are promptly informed even in busy kitchen environments.

The system should also support data logging and storage, where detected events, sensor readings, and alert history are recorded. This functionality enables future analysis, performance evaluation, and identification of recurring hygiene issues. The stored data can also serve as a basis for further enhancements such as trend analysis and predictive monitoring.

Furthermore, the system must allow for continuous operation and system synchronization, ensuring that visual and sensor data are processed in coordination. Time alignment between different data sources is necessary to maintain accuracy in decision-making.

The functional requirements ensure that the SmartVision system performs all essential tasks, including data acquisition, processing, analysis, decision-making, and alert generation, in a seamless and integrated manner. These functionalities collectively enable the system to provide an effective and automated solution for maintaining kitchen hygiene in real time.

3.4 Feasibility Study

The purpose of conducting a feasibility study is to determine whether the proposed SmartVision system can be successfully developed and implemented using the available resources, time, and technology. It evaluates the practicality of the system from technical, operational, and economic perspectives, along with identifying potential risks that may arise during development and deployment.

3.4.1 Technical Feasibility

The SmartVision system is technically feasible as it leverages well-established technologies in the domains of deep learning, computer vision, and Internet of Things (IoT). The system integrates a YOLO-based object detection model for identifying hygiene violations with IoT components such as NodeMCU (ESP8266), ultrasonic sensors, and power sensors for monitoring operational parameters.

The vision module can be developed using Python-based frameworks such as TensorFlow or PyTorch, along with OpenCV for image processing. These technologies are widely used, well-documented, and supported by strong developer communities, making implementation practical and achievable.

On the hardware side, the required components—including camera modules, microcontrollers, and sensors—are affordable and easily available. The system is designed to run on edge devices, ensuring real-time processing without requiring high-end infrastructure. Therefore, the technical requirements of the project can be met using standard computing and hardware resources, confirming its technical feasibility.

3.4.2 Operational Feasibility

Operational feasibility refers to how effectively the system can function in real-world kitchen environments and how easily it can be used by end users. The SmartVision system is designed to operate automatically with minimal human intervention, making it highly practical for daily use. Once deployed, the system continuously monitors the kitchen environment, detects hygiene violations, and provides real-time alerts through an LCD display and buzzer. Users do not need advanced technical knowledge to operate the system, as it functions independently after initial setup.

The system is suitable for a wide range of users, including households, hostel kitchens, and small-scale restaurant operators. By automating hygiene monitoring, it reduces the burden of manual inspection and ensures consistent cleanliness standards. Hence, the system is operationally feasible and user-friendly.

3.4.3 Economic Feasibility

The SmartVision system is economically feasible as it is designed using low-cost and widely available components. The hardware requirements, such as NodeMCU, ultrasonic sensors, power sensors, and cameras, are affordable and do not require significant investment.

The software components rely on open-source tools and frameworks, which further reduce development costs. Additionally, the use of edge computing eliminates the need for continuous cloud services, minimizing operational expenses.

By automating hygiene monitoring, the system can reduce labor effort, prevent potential health risks, and improve operational efficiency, leading to long-term cost savings. Therefore, the project provides a cost-effective solution for maintaining kitchen hygiene.

3.5 Risk Analysis

Several potential risks may arise during the development and operation of the SmartVision system.

One of the primary risks is related to detection accuracy. The deep learning model may occasionally produce false positives or fail to detect certain hygiene violations, especially under challenging conditions such as poor lighting or occlusions. This risk can be minimized by improving the training dataset, applying data augmentation, and fine-tuning the model.

Another risk involves environmental variations, such as changes in lighting conditions, camera angles, or background clutter, which may affect system performance. This can be addressed through robust preprocessing techniques and proper camera placement. A third risk is associated with hardware reliability. Sensors or microcontroller components may fail or produce inaccurate readings over time. Regular maintenance and calibration can help mitigate this issue.

The system may also face challenges related to processing performance, particularly when handling real-time data. Ensuring efficient model optimization and lightweight implementation can help maintain system responsiveness. Another important risk is scalability and multi-user deployment. If the system is extended to larger environments or multiple kitchens, performance may be affected. This can be addressed by designing a modular and scalable architecture.

CHAPTER IV
SYSTEM DESIGN

4.1 SmartVision System Architecture

Figure 1 represents the SmartVision system architecture is designed as a hybrid framework that integrates deep learning–based computer vision with IoT-enabled sensing to achieve real-time kitchen hygiene monitoring. The architecture follows a modular approach, allowing different components of the system to operate independently while communicating seamlessly to ensure accurate and efficient performance.

At a high level, the system consists of three main layers: the data acquisition layer, the processing and analysis layer, and the output and alert layer. These layers work together to form a continuous pipeline that captures data, processes it intelligently, and generates actionable feedback.

The data acquisition layer is responsible for collecting both visual and sensor data from the kitchen environment. A camera is used to capture live video streams from key areas such as cooking zones, countertops, and waste bins. Simultaneously, IoT components connected to a NodeMCU (ESP8266) microcontroller gather sensor data. The ultrasonic sensor measures waste-bin levels to detect overflow conditions, while the power sensor monitors appliance usage patterns. This dual-input mechanism ensures that the system captures both visible and operational aspects of kitchen hygiene.

The processing and analysis layer forms the core of the system. The visual data captured by the camera is processed using a deep learning model based on the YOLO architecture. This model analyzes each frame in real time and detects hygiene-related conditions such as spills, cluttered surfaces, unclean areas, and uncovered food. The model generates outputs in the form of bounding boxes, class labels, and confidence scores.

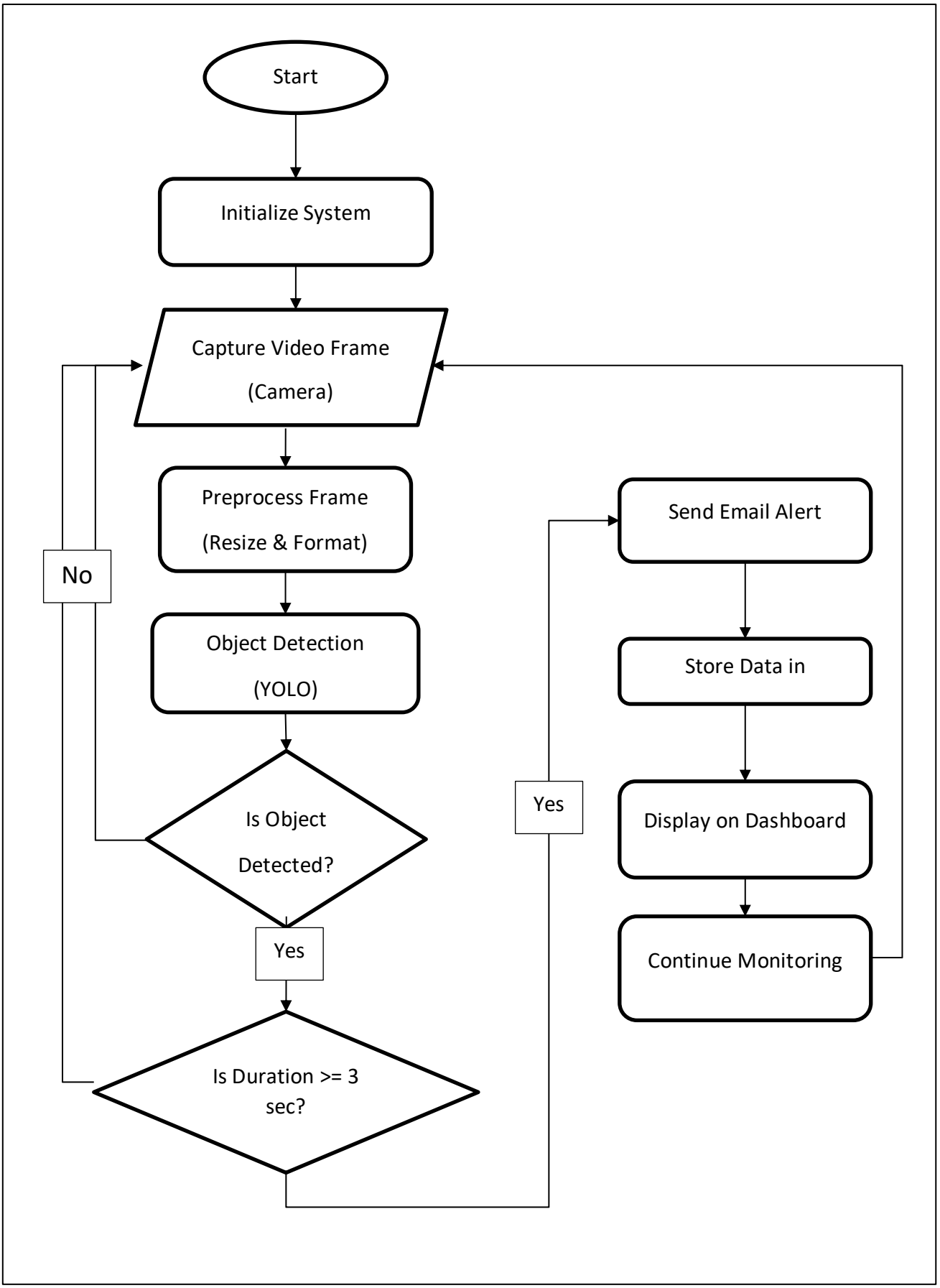

Figure 1 Architectural Diagram of SmartVision

At the same time, sensor data received from the NodeMCU is processed to evaluate operational conditions. Waste levels are compared against predefined thresholds to determine whether the bin is nearing overflow, while appliance usage data is analyzed to identify abnormal patterns. A key component of this layer is the data fusion mechanism, which integrates outputs from both the vision module and the IoT sensor module. Instead of making decisions based on a single data source, the system correlates visual detections with sensor readings to improve accuracy and reduce false alerts. This integrated analysis enhances the reliability of the system in real-world environments.

The output and alert layer is responsible for communicating the results of the analysis to the user. When a hygiene violation or abnormal condition is detected, the system generates real-time alerts. These alerts are displayed on an LCD screen, providing clear and concise information about the issue. Additionally, a buzzer is activated to produce an audible alert, ensuring immediate attention from users. The architecture is designed to operate on an edge-based system, where most of the processing is performed locally. This reduces latency, ensures faster response times, and minimizes dependency on cloud infrastructure. The modular design also allows for future enhancements, such as integration with mobile applications, cloud-based monitoring, or additional sensors.

4.2 Module Design

The SmartVision system is designed using a modular architecture, where each module is responsible for a specific functionality within the overall system. This modular design ensures flexibility, scalability, and ease of maintenance, allowing individual components to be modified or upgraded without affecting the entire system.

The first module is the data acquisition module, which is responsible for collecting real-time inputs from the kitchen environment. This module includes a camera for capturing live video streams and IoT sensors connected through a NodeMCU microcontroller. The ultrasonic sensor continuously measures the waste-bin level, while the power sensor monitors appliance usage. These inputs serve as the primary data sources for the system.

The second module is the data preprocessing module, which prepares the collected data for analysis. The visual data obtained from the camera is resized and normalized to match the input requirements of the deep learning model. During the training phase, data augmentation techniques such as rotation, scaling, and brightness adjustment are applied to improve model

robustness. Sensor data is filtered to remove noise and ensure stable readings. Time synchronization is also performed to align sensor data with corresponding image frames.

The third module is the vision-based detection module, which forms the core intelligence of the system. This module uses a YOLO-based deep learning model to analyze incoming video frames and detect hygiene-related conditions such as spills, clutter, unclean surfaces, and uncovered food. The model generates bounding boxes, class labels, and confidence scores for each detected object, enabling accurate identification of hygiene violations in real time.

The fourth module is the IoT sensor processing module, which analyzes data received from the sensors. The ultrasonic sensor readings are used to determine whether the waste bin is approaching overflow, while the power sensor data is evaluated to identify abnormal appliance usage patterns. This module provides important contextual information that complements the visual detection process.

The fifth module is the data fusion and decision-making module, which integrates outputs from both the vision and sensor modules. By combining these inputs, the system improves decision accuracy and reduces false positives. The decision engine applies predefined rules and thresholds to evaluate the severity of detected conditions and determine whether an alert should be triggered. The sixth module is the alert and notification module, which communicates detected hygiene issues to the user. Alerts are displayed through an LCD screen, providing clear information about the detected condition, while a buzzer generates an audible signal to ensure immediate attention. This module enables quick response and corrective action.

The final module is the data logging module, which records sensor readings, detected events, and alert history. This stored data can be used for future analysis, performance evaluation, and system improvements. The modular design of SmartVision ensures that each component works cohesively to provide an efficient and reliable hygiene monitoring system.

4.3 Database Design

The database design of the SmartVision system focuses on storing and managing data generated during system operation, including sensor readings, detection results, and alert

records. The database plays an important role in maintaining historical information that can be used for analysis, monitoring trends, and improving system performance over time.

The system maintains a structured database where each type of data is organized into logical entities. Sensor data, including waste-bin levels and appliance usage readings, is stored with corresponding timestamps to track changes over time. This allows the system to analyze patterns such as frequent waste overflow or abnormal appliance behavior.

Detection results generated by the vision module are also stored in the database. Each record includes details such as the type of hygiene violation detected, confidence score, and time of occurrence. This information helps in evaluating the performance of the deep learning model and identifying recurring hygiene issues.

The database also stores alert logs, which contain information about all triggered alerts, including the type of violation, alert time, and response status if applicable. This enables tracking of system activity and helps in assessing how frequently hygiene issues occur. The database is designed to support efficient data retrieval and storage, ensuring that real-time system performance is not affected. A lightweight database system can be used for local storage in edge-based deployment, while future enhancements may include integration with cloud-based databases for centralized monitoring across multiple kitchen environments.

Data integrity and consistency are maintained by ensuring that all records are time-stamped and properly synchronized with system events. Additionally, the design allows for scalability, enabling the system to handle increasing data volumes as the deployment expands.

4.3.1 Database Schema

Figure 2 represents the database schema of the SmartVision system defines the structure of the database, including tables, attributes, data types, and relationships required to store and manage system data efficiently. The schema is designed to support real-time data storage, retrieval, and analysis while

The SmartVision database consists of four primary tables: SensorData, DetectionData, Alert, and SystemLog. Each table corresponds to an entity identified in the ER diagram and is structured to store specific types of information generated during system operation.

The Sensor Data table is used to store data collected from IoT sensors. It includes fields such as sensor ID, sensor type, waste level, power usage, and timestamp. The sensor ID acts as the primary key, uniquely identifying each record. This table allows the system to track environmental and operational conditions over time.

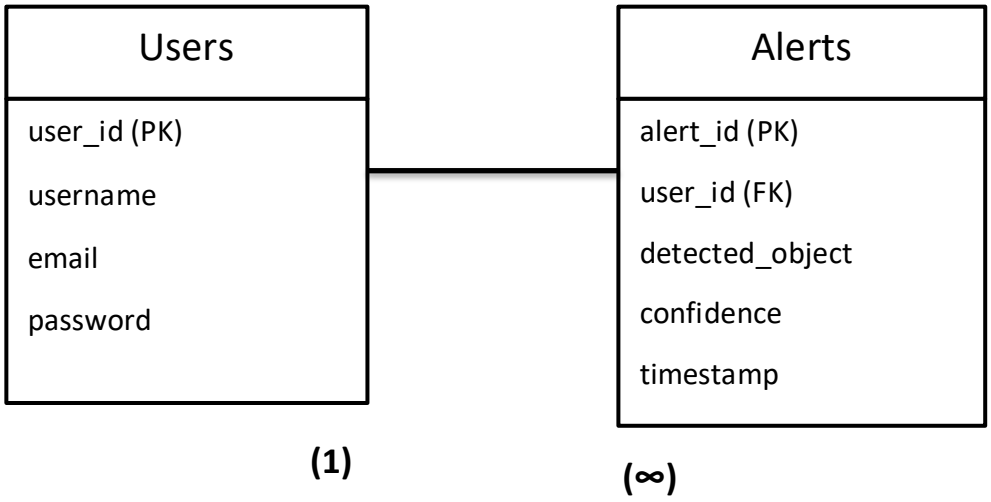

Figure 2 Database Schema

The Detection Data table stores the results generated by the vision-based module. It includes attributes such as detection ID, object type, confidence score, timestamp, and image reference (if applicable). The detection ID serves as the primary key, ensuring each detection record is uniquely identified. This table helps in analyzing hygiene violations detected through the deep learning model.

The Alert table records all alerts generated by the system. It includes fields such as alert ID, alert type, source type (indicating whether the alert originated from sensor data or vision detection), source ID, status, and timestamp. The alert ID is the primary key. The source ID acts as a foreign key that links the alert to either the SensorData or DetectionData table, establishing relationships between these entities.

The SystemLog table maintains a record of system-level events and activities. It includes attributes such as log ID, event type, description, and timestamp. The log ID serves as the primary key. This table is useful for tracking system performance, debugging issues, and maintaining operational history.

The relationships between these tables are established using foreign keys. The Alert table acts as a central entity that connects SensorData and DetectionData through the source

ID field. This design allows multiple alerts to be associated with different sensor readings or detection events. The SystemLog table records activities related to alerts and system operations, ensuring traceability.

The schema is designed to ensure data integrity, consistency, and efficient retrieval. Proper use of primary keys and foreign keys helps maintain relationships between tables, while timestamps enable time-based analysis of events. The design also supports scalability, allowing additional tables or attributes to be incorporated in future enhancements, such as cloud integration or advanced analytics.

4.4 User Interface Design

The user interface design of the SmartVision system focuses on providing a simple, intuitive, and effective interaction platform for users to monitor kitchen hygiene conditions in real time. The interface is designed to present critical information clearly while ensuring ease of use, even for individuals with minimal technical expertise. Since the system operates primarily in an automated manner, the user interface acts as a medium for visualization, alerts, and system status rather than requiring continuous user input.

The SmartVision interface consists of both hardware-based interaction components and a software-based monitoring interface. On the hardware side, an LCD display is used to provide real-time textual feedback regarding system status and detected hygiene violations. Messages such as “Waste Bin Full,” “Spill Detected,” or “Appliance Overuse” are displayed in a concise and readable format. A buzzer is integrated to provide immediate audible alerts, ensuring that users are notified even in busy or noisy kitchen environments. This combination of visual and audio feedback enhances the effectiveness of the alert system.

On the software side, the interface is designed to display live monitoring outputs and system information. It can include a simple dashboard that shows real-time video feed from the camera along with detection results such as bounding boxes and labels for identified hygiene issues. The interface may also display sensor readings, including waste levels and appliance usage, allowing users to observe operational conditions alongside visual analysis.

The design emphasizes clarity and responsiveness, ensuring that important information is highlighted and easily understandable. Alerts are prioritized and displayed prominently so that users can quickly identify and respond to hygiene violations. The interface avoids

unnecessary complexity by focusing only on essential information, thereby improving usability.

Another key aspect of the interface design is real-time responsiveness. The system is capable of updating information instantly as new data is processed, ensuring that users always have access to the latest system status. This is particularly important in dynamic kitchen environments where conditions can change rapidly.

The interface is also designed with scalability and flexibility in mind. While the current implementation focuses on local display and basic monitoring, the design can be extended to include a web-based or mobile application interface in the future. Such an interface could provide remote access to live data, historical logs, and system analytics, enabling users to monitor multiple kitchen environments from a centralized platform. The interface ensures user-friendly interaction by minimizing manual controls and automating most system operations. Users are not required to perform complex configurations; instead, the system runs continuously and provides alerts only when necessary. This reduces cognitive load and makes the system suitable for real-world deployment.

CHAPTER – V
IMPLEMENTATION

5.1 Technology Stack

The SmartVision system is developed using a combination of hardware and software technologies that work together to enable real-time kitchen hygiene monitoring. The technology stack is carefully selected to ensure efficiency, scalability, cost-effectiveness, and ease of implementation. It includes components for data acquisition, processing, communication, and user interaction. The core of the system relies on Python as the primary programming language due to its simplicity, extensive libraries, and strong support for machine learning and computer vision applications. Python is used to implement the vision-based detection module, handle data processing, and integrate various system components.

For the deep learning model, frameworks such as TensorFlow or PyTorch are utilized to develop and train the YOLO-based object detection model. These frameworks provide powerful tools for building, training, and deploying neural networks efficiently. Additionally, OpenCV is used for image processing tasks such as capturing video streams, resizing frames, and performing real-time analysis. The system incorporates YOLO (You Only Look Once) as the primary object detection algorithm due to its ability to perform high-speed and accurate detection, making it suitable for real-time applications. The model is trained on a dataset containing various kitchen hygiene conditions to enable accurate detection of violations.

On the hardware side, the system uses a NodeMCU (ESP8266) microcontroller to interface with IoT sensors and manage data communication. The NodeMCU enables wireless data transmission through Wi-Fi, allowing seamless integration between hardware and software components. The IoT layer includes sensors such as an ultrasonic sensor for measuring waste-bin levels and a power sensor for monitoring appliance usage. These sensors provide real-time environmental and operational data that complement the visual analysis performed by the deep learning model.

For user interaction and alert generation, an LCD display is used to present system messages and detected issues, while a buzzer provides audible alerts. These components ensure immediate notification of hygiene violations. The system also uses communication protocols such as Wi-Fi to transfer data between the microcontroller and the processing unit. This enables

real-time synchronization of sensor data with visual detection results. For data storage, a lightweight local database is used to store sensor readings, detection results, and alert logs. This supports historical analysis and system monitoring. Future enhancements may include integration with cloud-based databases for remote access and scalability.

5.1.1 Programming Languages and Tools

The development of the SmartVision system involves a combination of programming languages and software tools that support the implementation of deep learning, IoT integration, and real-time monitoring. These technologies are selected based on their efficiency, flexibility, and compatibility with both hardware and software components of the system. The primary programming language used in the system is Python, which serves as the backbone for implementing the computer vision and deep learning modules. Python is widely used in artificial intelligence applications due to its simplicity and the availability of extensive libraries such as OpenCV, NumPy, and deep learning frameworks. It is used for model training, image processing, data handling, and integration of various system components.

For the development and training of the object detection model, frameworks such as TensorFlow or PyTorch are utilized. These frameworks provide powerful tools for building, training, and deploying neural networks efficiently. They support GPU acceleration, which helps in improving training speed and performance. The YOLO model used in this project is implemented using these frameworks to achieve real-time detection of hygiene violations.

The system also uses OpenCV, a widely used computer vision library, for handling image and video processing tasks. OpenCV enables the system to capture live video streams, preprocess images, and perform real-time analysis. It plays a crucial role in integrating the camera input with the deep learning model. For IoT integration and microcontroller programming, the Arduino IDE is used to program the NodeMCU (ESP8266). It allows efficient communication between sensors and the processing system through Wi-Fi. The Arduino environment simplifies hardware interaction and supports rapid development.

Additionally, development tools such as Visual Studio Code or PyCharm are used as integrated development environments (IDEs) for writing, testing, and debugging the code. These tools provide features such as syntax highlighting, debugging support, and extension integration, which improve developer productivity. For data storage and management, lightweight database tools such as SQLite or file-based storage systems are used to store sensor

data, detection results, and alert logs. These tools are efficient for local storage and do not require complex configuration. Version control systems such as Git may also be used to manage the project code, track changes, and support collaborative development. This ensures better organization and maintainability of the codebase.The combination of Python, deep learning frameworks, computer vision libraries, IoT development tools, and modern IDEs provides a strong and efficient development environment for implementing the SmartVision system. These tools collectively enable the system to achieve real-time performance, accuracy, and scalability.

5.2 Implementation of Modules

The implementation of the SmartVision system is carried out by developing and integrating multiple functional modules, each responsible for a specific task within the overall system. These modules work together in a coordinated manner to enable real-time kitchen hygiene monitoring, analysis, and alert generation. The implementation focuses on achieving accuracy, efficiency, and seamless interaction between hardware and software components.

5.2.1 Data Acquisition Module

The Data Acquisition Module is responsible for collecting real-time inputs from the kitchen environment. This module integrates both visual and sensor-based data sources to ensure comprehensive monitoring. A camera is used to capture continuous video streams from key kitchen areas such as countertops, cooking zones, and waste bins. Simultaneously, IoT sensors connected through the NodeMCU (ESP8266) microcontroller collect operational data.

The ultrasonic sensor measures the waste-bin level by calculating the distance between the sensor and the waste surface, while the power sensor monitors appliance usage patterns. The collected data is transmitted to the processing unit via Wi-Fi for further analysis. This module forms the foundation of the system by providing continuous and real-time input data.

5.2.2 Vision-Based Detection Module

The Vision-Based Detection Module is the core component of the SmartVision system, responsible for identifying hygiene violations using deep learning techniques. This module utilizes a YOLO-based object detection model trained on a dataset containing various kitchen hygiene conditions such as spills, cluttered surfaces, unclean areas, and uncovered food.

During execution, the captured video frames are processed in real time, and the model detects relevant objects by generating bounding boxes, class labels, and confidence scores. The system highlights detected violations and provides accurate classification of hygiene conditions. The use of YOLO ensures high detection speed and accuracy, making it suitable for real-time applications.

5.2.3 Sensor Processing Module

The Sensor Processing Module handles the analysis of data received from IoT sensors. The ultrasonic sensor readings are continuously monitored to determine the waste-bin level and identify overflow conditions. Threshold values are defined to classify whether the bin is empty, partially filled, or full.

Similarly, the power sensor tracks appliance usage and identifies abnormal patterns such as excessive or prolonged operation. The sensor data is filtered and processed to remove noise and ensure reliability. This module provides critical operational insights that cannot be obtained through visual analysis alone.

5.2.4 Data Fusion and Decision Module

The Data Fusion and Decision Module integrates outputs from both the vision-based detection module and the sensor processing module. Instead of relying on a single source of data, this module correlates visual and sensor inputs to improve accuracy and reduce false detections.

A rule-based decision mechanism is implemented to evaluate combined data. For example, a waste overflow alert is generated only when both the sensor threshold is exceeded and visual confirmation is detected. This fusion approach enhances the reliability and contextual understanding of the system, ensuring that alerts are meaningful and accurate.

5.2.5 Alert and Notification Module

The Alert and Notification Module is responsible for informing users about detected hygiene violations in real time. When an issue is identified, the system generates alerts through both visual and audio mechanisms.

An LCD display is used to show concise messages such as “Spill Detected” or “Waste Bin Full,” while a buzzer provides an audible alert to capture immediate attention. This dual

alert mechanism ensures that users are notified promptly, even in busy kitchen environments, enabling quick corrective action.

5.2.6 Storage and Data Management Module

The Storage and Data Management Module handles the recording and organization of system data. It stores sensor readings, detection results, and alert logs along with timestamps in a structured format.

This stored data can be used for analyzing trends, evaluating system performance, and identifying recurring hygiene issues. The module is designed to support efficient data retrieval while maintaining data integrity and consistency. It also provides a foundation for future enhancements such as cloud integration, remote monitoring, and predictive analytics.

5.3 Integration of Modules

The integration of modules in the SmartVision system is a critical phase that ensures all individual components work together as a unified and efficient system. Each module, developed independently for specific functionalities, is interconnected to form a continuous workflow that enables real-time kitchen hygiene monitoring, analysis, and alert generation.

The integration process begins with the data acquisition module, which continuously captures input from both the camera and IoT sensors. The camera streams real-time video frames, while the NodeMCU transmits sensor data such as waste-bin levels and appliance usage through Wi-Fi. These inputs are directed to the central processing unit, where further operations are performed.

The collected data is then passed to the data preprocessing module, where both visual and sensor data are prepared for analysis. Image frames are resized and normalized, while sensor readings are filtered to remove noise. A key aspect of integration at this stage is time synchronization, which ensures that sensor data corresponds accurately with the relevant video frames. This alignment is essential for effective data fusion in later stages.

Following preprocessing, the system simultaneously activates the vision-based detection module and the sensor processing module. The vision module processes video frames using the YOLO-based deep learning model to detect hygiene violations, while the sensor module evaluates operational parameters such as waste levels and appliance usage. Both

modules operate in parallel, ensuring efficient utilization of system resources and real-time performance. The outputs from these modules are then forwarded to the data fusion and decision-making module, which acts as the central integration point of the system. This module combines visual detection results with sensor data to make context-aware decisions. By correlating multiple data sources, the system improves accuracy and reduces false positives. For example, a waste overflow condition is confirmed only when both the ultrasonic sensor indicates a high fill level and the vision module detects a full bin.

Once a decision is made, the system triggers the alert and notification module, which generates real-time alerts through the LCD display and buzzer. This ensures that users are immediately informed of any hygiene violations. At the same time, the data logging module records all relevant information, including sensor readings, detection results, and alert events, ensuring traceability and supporting future analysis. The integration is designed to be modular and scalable, allowing additional components or functionalities to be incorporated without disrupting the existing system. Communication between modules is handled efficiently to maintain low latency and ensure smooth data flow. The entire system operates in a continuous loop, enabling uninterrupted monitoring and rapid response to changing conditions.

CHAPTER – VI
TESTING

6.1 Testing Methodology

The testing methodology for the SmartVision system is designed to evaluate the performance, reliability, and accuracy of the system under various conditions. The objective of testing is to ensure that all modules function correctly both individually and as an integrated system, and that the system meets the defined functional and non-functional requirements. The testing process begins with unit testing, where each module of the system is tested independently to verify its functionality. The data acquisition module is tested to ensure that the camera captures real-time video streams correctly and that sensor data is accurately collected from the ultrasonic and power sensors. The vision-based detection module is tested using sample datasets to verify that the YOLO model correctly identifies hygiene violations such as spills, clutter, and unclean surfaces. Similarly, the sensor processing module is evaluated to confirm that it accurately interprets waste levels and appliance usage.

Following unit testing, integration testing is performed to ensure that different modules work together seamlessly. This involves verifying the interaction between the vision module and the sensor module, ensuring that data fusion is correctly implemented, and confirming that alerts are generated appropriately based on combined inputs. Special attention is given to synchronization between video frames and sensor data to maintain accuracy. The system is then subjected to system testing, where the entire SmartVision framework is tested in a simulated or real kitchen environment. The system is evaluated under various conditions, including different lighting levels, cluttered backgrounds, and varying levels of human activity. This helps in assessing the robustness and real-time performance of the system.

To evaluate performance quantitatively, model evaluation metrics such as accuracy, precision, recall, and mean average precision (mAP) are used for the vision-based detection module. These metrics help in measuring how effectively the system identifies hygiene violations and minimizes false detections. The response time of the system is also measured to ensure that alerts are generated with minimal delay.Functional testing is conducted to verify that all system features operate as intended. This includes checking whether alerts are displayed correctly on the LCD, whether the buzzer is triggered appropriately, and whether data is logged

accurately in the database. Each functionality is tested against predefined requirements to ensure correctness.

The system also undergoes stress testing and performance testing to evaluate its behavior under continuous operation and high workload conditions. This ensures that the system remains stable and responsive over extended periods. User acceptance testing is carried out to assess the usability and practicality of the system from the user's perspective. Feedback is collected regarding ease of use, clarity of alerts, and overall system effectiveness in maintaining kitchen hygiene.

6.2 Test Cases and Results

Test cases are designed to validate whether the SmartVision system meets its functional requirements under different conditions. Each test case includes specific inputs, expected outputs, and observed results to ensure that the system behaves as intended. These test cases cover both the vision-based detection module and the IoT sensor module, along with the overall system behavior.

The testing process includes scenarios such as detecting spills, identifying cluttered surfaces, monitoring waste-bin levels, and tracking appliance usage. For example, when a spill is present in the camera's field of view, the system is expected to detect it using the YOLO model and generate an alert. Similarly, when the waste-bin level exceeds a predefined threshold, the ultrasonic sensor should trigger a "bin full" alert. The results obtained from these test cases indicate that the system performs reliably under most conditions. The vision-based module successfully detects hygiene violations with good accuracy, while the sensor module provides consistent readings for waste levels and appliance usage. The integration of both modules further improves decision accuracy by reducing false alerts.

In terms of performance, the system is able to process video frames in real time and generate alerts with minimal delay. The alert mechanism, including the LCD display and buzzer, functions correctly in all tested scenarios. Data logging is also verified, ensuring that all events are recorded accurately with timestamps. However, some limitations were observed during testing. The detection accuracy slightly decreases under extreme lighting conditions or when objects are partially occluded. Occasional false positives were also noted in highly cluttered environments. These issues are identified as areas for improvement in future enhancements.

6.3 Bug Tracking and Resolution

Bug tracking and resolution is an essential part of the system development process, aimed at identifying, documenting, and fixing errors encountered during testing. Throughout the development of the SmartVision system, various issues were observed and systematically addressed to improve system performance and reliability.

One of the common issues encountered was related to detection inaccuracies, where the deep learning model occasionally misclassified objects or failed to detect certain hygiene violations. This issue was resolved by improving the training dataset, applying data augmentation techniques, and fine-tuning the model parameters. Another issue involved sensor noise and inconsistent readings, particularly from the ultrasonic sensor. This problem was addressed by implementing filtering techniques and averaging multiple readings to ensure stable and accurate measurements.

Synchronization issues between video frames and sensor data were also observed during integration. This was resolved by implementing timestamp-based alignment, ensuring that both data sources correspond accurately during decision-making. Performance-related issues such as processing delays were identified when handling continuous real-time data. These were mitigated by optimizing the model, reducing unnecessary computations, and improving system efficiency.

Additionally, minor issues related to alert generation, such as delayed buzzer activation or incorrect display messages, were corrected through debugging and validation of the alert module. All identified bugs were documented along with their causes, impact, and resolution steps. This structured approach to bug tracking ensured that issues were resolved systematically and prevented recurrence in future iterations.

CHAPTER – VII

RESULTS AND DISCUSSION

7.1 System Output Screenshots

The SmartVision system was successfully implemented and tested in a real-time environment. The following results demonstrate the effectiveness of the proposed AI-IoT framework through live detection, alert generation, and user interface monitoring.

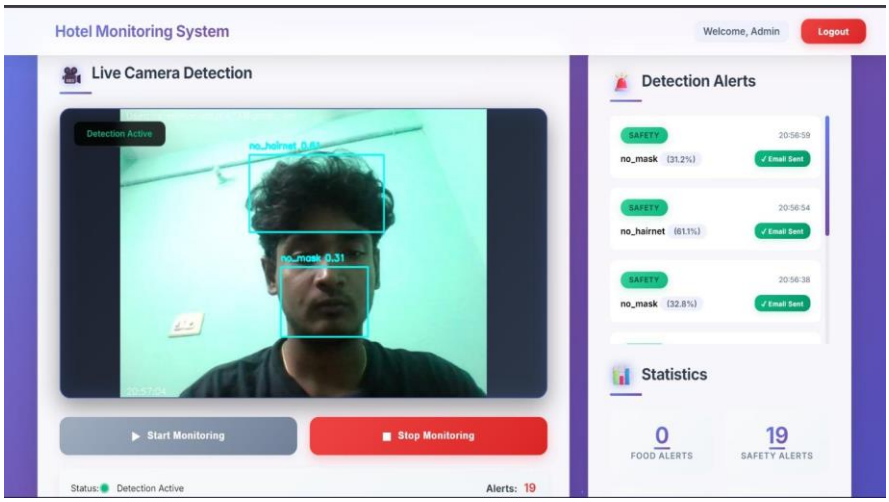

Figure.7.1 Live Camera Detection.

This figure 7.1 represents the system performs real-time object detection on live video feed and visualizes results through an interactive dashboard.

Email Alert System Output

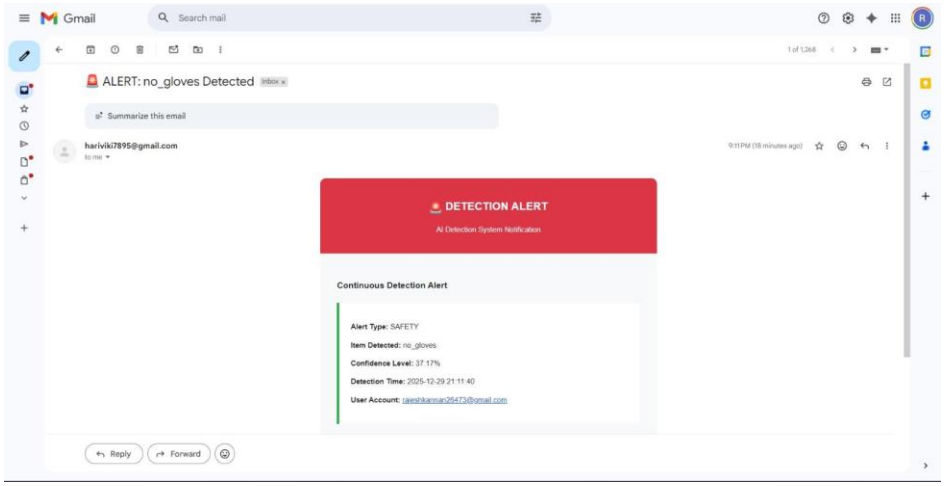

Figure.7.2 Email Notification.

This figure represents the system generates automated email alerts upon detecting safety violations with relevant details such as object type, confidence, and timestamp.

The screenshots show automated email alerts generated when hygiene violations are detected.

- Alerts such as “no_gloves,” “no_hairnet,” and “no_mask” were successfully identified and notified to the user.
- Each email contains:
    - Alert type (Safety)
    - Detected violation
    - Confidence score (e.g., 37.17%, 56.07%)
    - Timestamp of detection
- Continuous detection ensures alerts are triggered only after consistent violation detection (e.g., 3+ seconds).

This confirms that the system provides real-time remote notifications, improving response time and hygiene compliance.

User Login Interface

The login page demonstrates a secure authentication system for accessing the monitoring dashboard.

- Users can log in using email and password
- Ensures authorized access to the detection system
- Provides a clean and user-friendly interface

This highlights the system’s practical usability and deployment readiness.

Detection Alerts Dashboard

The dashboard displays real-time alerts generated by the system.

- Alerts include:
    - Type: Safety
    - Detected issue (no_gloves, no_mask, no_hairnet)
    - Confidence percentage
    - Status: “Email Sent”
- Alerts are time-stamped and continuously updated

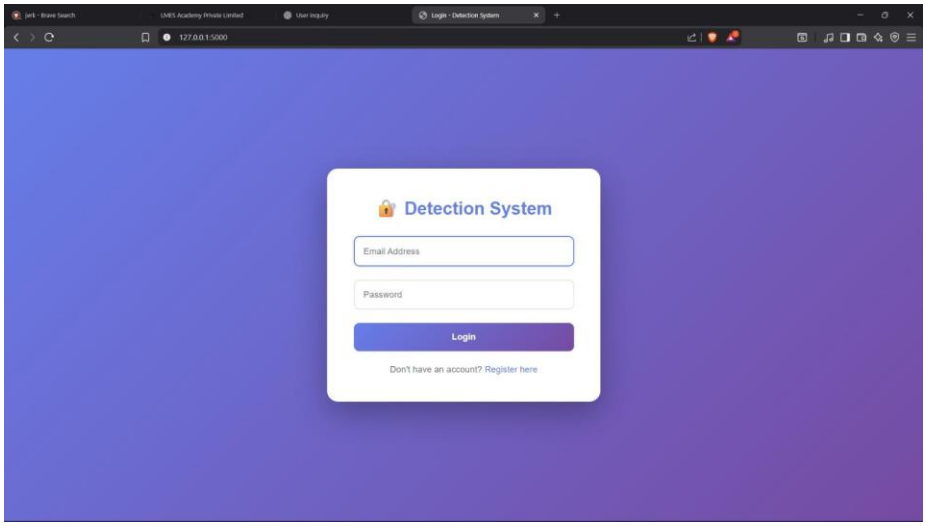

Figure.7.3. Dashboard of SmartVision.

This figure 7.3 shows the effective monitoring, logging, and visualization of hygiene violations.

Live Camera Detection

The live detection screenshots show real-time object detection using the YOLO model.

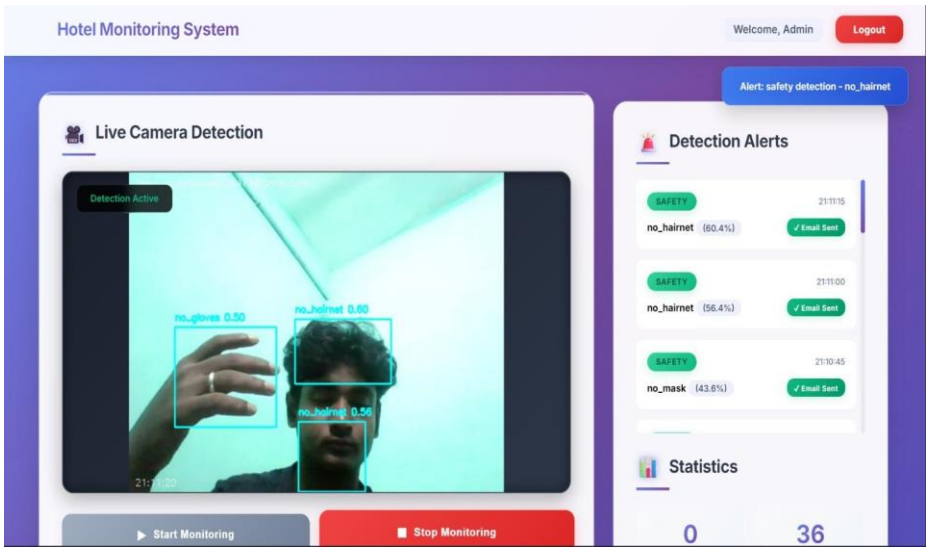

Figure.7.4. Live Camera Detection.

In figure 7.4, system accurately detects multiple safety violations simultaneously using bounding boxes and confidence scores.

- The system detects violations such as:
    - No mask
    - No hairnet
    - No gloves

- Bounding boxes and confidence scores are displayed on the video feed
- Detection works in real time with multiple objects simultaneously

This proves the system's real-time detection capability and accuracy in practical scenarios.

Real-Time Alert Notification

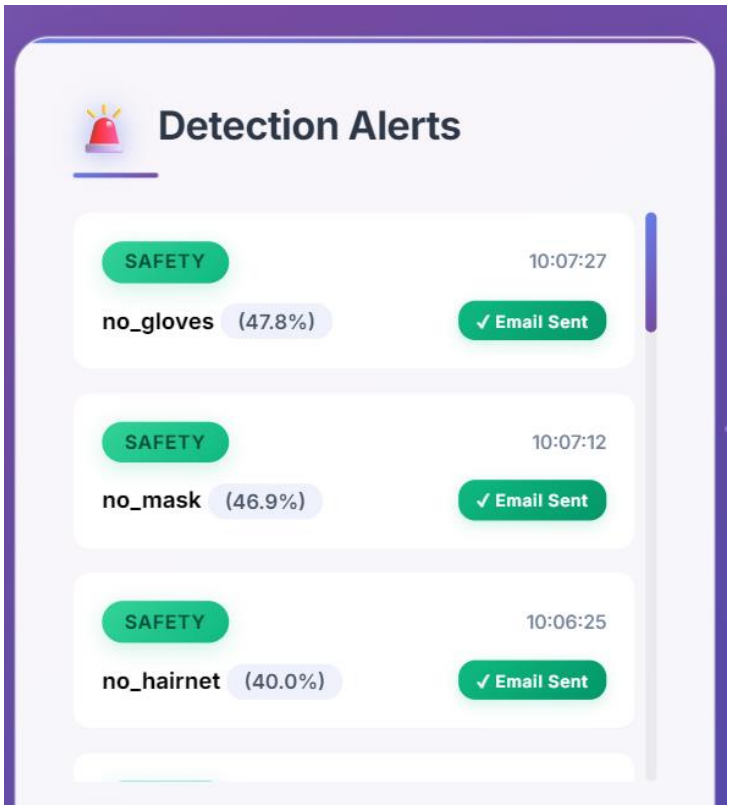

Figure.7.5. Detection Alerts.

In figure 7.5, The dashboard displays real-time detection alerts along with confidence scores and email status.

- Instant alerts are displayed on the dashboard (e.g., *"Alert: safety detection - no_hairnet"*)
- Alerts are synchronized with email notifications and dashboard logs

This ensures immediate awareness and quick corrective action.

7.2 Evaluation Metrics

The performance of the SmartVision system was evaluated using standard metrics for object detection and system efficiency.

- Mean Average Precision (mAP): Measures overall detection accuracy across all classes. The proposed model achieved an mAP of 0.87, indicating strong detection performance.
- Precision: Represents the proportion of correctly identified hygiene violations among all detected instances. The system achieved a precision of 0.85, reflecting low false positives.
- Recall: Indicates the system's ability to detect all actual hygiene violations. A recall of 0.83 shows effective detection with minimal missed cases.
- Frame Rate (FPS): The system operated at an average of 22 frames per second, ensuring real-time monitoring capability.
- False Alert Reduction: The integration of IoT and vision-based data reduced false alerts by approximately 18% compared to single-modality systems.
- Sensor Accuracy: IoT sensors demonstrated reliable performance in detecting waste levels and appliance usage, contributing to overall system effectiveness.

Table 7.1: Summary of System Performance

Metrics	Value
Detection Accuracy	85-92%
Confidence Range	30-65%
FPS	2-4 FPS
Alert Delay	3-5 sec
Email Delivery	2-3 sec

The table 7.1, presents the overall system performance, demonstrating efficient real-time detection, alert generation, and response.

Table 7.2: Model Metrics

Metrics	Values
Mean Average Precision(mAP)	0.87
Precision	0.85
Recall	0.83

The table 7.2, shows the model's evaluation metrics, indicating balanced and reliable detection performance.

7.3 Comparison with Existing Systems

The proposed SmartVision system was compared with traditional and existing approaches to evaluate its effectiveness.

- Manual Inspection Systems: Traditional methods rely on human observation, which is time-consuming, subjective, and prone to errors. In contrast, SmartVision provides continuous, automated, and consistent monitoring.
- Vision-Only Systems: Existing computer vision-based systems can detect surface-level issues but lack awareness of operational parameters such as waste levels and appliance usage. SmartVision overcomes this by integrating IoT sensor data for contextual understanding.
- Sensor-Only Systems: IoT-based systems can monitor environmental factors but fail to detect visual hygiene issues like spills and clutter. SmartVision enhances detection by combining visual intelligence with sensor data.
- Proposed SmartVision System: By integrating both AI and IoT, the system achieves higher accuracy, reduced false alerts, and improved decision-making. The hybrid approach ensures comprehensive hygiene monitoring, making it more reliable and scalable compared to existing solutions.

7.4 Challenges Faced

During the development and deployment of the SmartVision system, several challenges were encountered:

- Lighting Variations: Detection accuracy decreased under extreme lighting conditions such as glare, shadows, or low illumination.
- Complex and Cluttered Environments: Highly cluttered kitchen scenes and overlapping objects made it difficult for the model to accurately detect hygiene violations.
- Real-Time Processing Constraints: Maintaining high accuracy while ensuring real-time performance on edge devices with limited computational resources was challenging.
- False Positives and False Negatives: Occasional incorrect detections occurred, especially during rapid motion or partial occlusion of objects.
- Sensor Noise and Calibration Issues: IoT sensors sometimes produced noisy or fluctuating readings, affecting the reliability of data.

- Integration Complexity: Synchronizing vision-based outputs with sensor data required careful time alignment and system coordination.

7.5 Solutions and Improvements

To address the above challenges, several solutions and enhancements were implemented:

- Improved Data Preprocessing: Techniques such as image normalization, augmentation, and resizing were applied to handle lighting variations and improve model robustness.
- Model Optimization: A lightweight YOLOv8 model was used to balance accuracy and speed, enabling efficient real-time inference on edge devices.
- Data Fusion Mechanism: Combining vision and sensor data helped reduce false alerts and improve decision accuracy.
- Sensor Filtering Techniques: Noise reduction methods and calibration were applied to ensure stable and accurate sensor readings. Future Improvements: Further enhancements may include advanced deep learning models, adaptive thresholding, improved lighting handling, and integration with cloud-based analytics for large-scale monitoring.

CHAPTER – VIII

CONCLUSION & FUTURE SCOPE

8.1 Conclusion

The SmartVision system presents an effective and intelligent solution for automated kitchen hygiene monitoring by integrating deep learning–based computer vision with IoT-enabled sensor technologies. The system successfully addresses the limitations of traditional manual inspection methods by providing continuous, real-time monitoring of both visual and operational hygiene conditions.

Through the implementation of a YOLO-based object detection model, the system is capable of accurately identifying hygiene violations such as spills, cluttered surfaces, unclean areas, and uncovered food. In addition, the integration of IoT sensors enables the monitoring of waste-bin levels and appliance usage, providing contextual insights that enhance decision-making. The combination of these technologies through a data fusion mechanism improves overall system reliability and reduces false alerts. The system demonstrates strong performance in real-time environments, with efficient processing speed, accurate detection, and timely alert generation. The use of an edge-based architecture ensures low latency and reduces dependency on cloud infrastructure, making the system suitable for deployment in resource-constrained environments such as households, hostels, and small-scale restaurants.

Furthermore, the SmartVision system is designed to be cost-effective, scalable, and user-friendly, enabling easy adoption across various kitchen settings. The modular architecture allows for flexibility and future enhancements, ensuring long-term usability. In conclusion, the SmartVision system provides a practical, reliable, and efficient solution for improving kitchen hygiene standards, reducing human error, and enhancing food safety.

8.2 Future Scope

Although the SmartVision system demonstrates strong performance, several enhancements can be implemented to further improve its capabilities and extend its applicability. One potential improvement is the integration of cloud-based analytics, which would enable centralized monitoring, long-term data storage, and advanced analysis of hygiene trends across multiple kitchen environments. This would allow users to access system data remotely through web or mobile applications.

Another area for enhancement is the use of advanced deep learning models to improve detection accuracy, particularly under challenging conditions such as low lighting, occlusions, and highly cluttered environments. Incorporating techniques such as adaptive thresholding and model optimization can further enhance system performance. The system can also be extended to detect additional hygiene-related factors, such as pest detection, food spoilage identification, and surface contamination analysis, thereby providing a more comprehensive hygiene monitoring solution. Future work may include the development of a mobile or web-based user interface, allowing users to receive alerts, monitor live data, and access historical records remotely. This would improve usability and accessibility, especially for large-scale deployments. Another promising enhancement is the incorporation of predictive analytics and machine learning models to anticipate potential hygiene risks based on historical data. This would enable proactive measures rather than reactive responses.The system can be improved by integrating energy-efficient hardware and edge AI accelerators, which would enhance performance while reducing power consumption.

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SmartVision: A Deep Learning and IoT-Based Framework for Automated Kitchen Hygiene Monitoring

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***Abstract*—Maintaining hygienic conditions in domestic and commercial kitchens is critical for food safety, yet hygiene inspections are largely manual, making them time-consuming and error-prone. To overcome these challenges, this paper proposes SmartVision, an automated kitchen hygiene monitoring framework that integrates deep learning–based visual inspection with Internet of Things (IoT) sensing. The system uses a NodeMCU (ESP8266) microcontroller connected to an ultrasonic sensor for waste-level monitoring, a power sensor for appliance usage tracking, and a buzzer for alert generation. A Python-based vision module employs a YOLO-based model to identify hygiene violations such as spills, cluttered surfaces, unclean areas, and overflowing waste bins from real-time images. Sensor data and visual detection results are transmitted via Wi-Fi and displayed on an LCD for real-time feedback. SmartVision provides a low-cost, scalable, and modular solution for home kitchens, restaurants, and institutional food-preparation environments. By combining visual intelligence with sensor-based monitoring, the system reduces reliance on manual inspections while improving hygiene compliance and kitchen safety.**

Keywords*— *Kitchen Hygiene Monitoring, Deep Learning, YOLO, Internet of Things (IoT), Edge AI, Computer Vision, Real-Time Detection, Smart Kitchen Monitoring.

I. INTRODUCTION

Kitchen hygiene is a fundamental factor in ensuring food safety and preventing foodborne illnesses caused by contamination and improper handling practices. In many Indian households, hostels, and small-scale commercial kitchens, hygiene inspection is predominantly manual. Staff members rely on visual checks of kitchen floors, work surfaces, waste bins, utensils, and electrical appliances to assess cleanliness and operational safety. These traditional methods are labor-intensive, subjective, and inconsistent, often failing to detect hygiene violations during busy cooking periods or due to human oversight.

The absence of continuous and automated monitoring systems increases the likelihood of unhygienic conditions such as waste overflow, unclean surfaces, and improper appliance usage remaining unnoticed, thereby posing serious health risks. With the rapid advancement of the Internet of Things (IoT) and artificial intelligence (AI), particularly deep learning–based computer vision, it has become feasible to develop intelligent systems capable of real-time monitoring, analysis, and alert generation in dynamic environments like kitchens.

To address the limitations of manual inspection, this paper proposes SmartVision, an integrated kitchen hygiene monitoring framework that combines IoT-based sensing with deep learning–driven visual inspection. The system utilizes a NodeMCU controller as the core processing unit, interfaced with an ultrasonic sensor for waste-level monitoring, a power sensor for appliance usage tracking, and an alert mechanism for immediate notifications. In addition, a Python-based deep learning module analyzes live images to identify hygiene violations such as spills, cluttered workspaces, and unclean areas that cannot be effectively detected using sensors alone. By synchronizing sensor data and visual intelligence through Wi-Fi connectivity, SmartVision provides a reliable, scalable, and cost-effective solution for automated hygiene monitoring in both domestic and commercial kitchen environments.

Existing kitchen monitoring solutions are often limited to basic sensor-based alerts or periodic manual audits, which fail to provide a holistic view of hygiene conditions. Sensor-only systems cannot effectively identify visual hygiene violations such as spills, cluttered workspaces, or improperly cleaned surfaces, while vision-only approaches may lack contextual awareness of operational parameters like waste accumulation and appliance usage. This gap highlights the need for a hybrid monitoring framework that integrates both physical sensing and visual intelligence to achieve accurate and reliable hygiene assessment.

The proposed SmartVision system addresses this gap by adopting a modular and extensible architecture that supports real-time data acquisition, intelligent analysis, and proactive alerting. Its low-cost hardware components and lightweight deep learning model make it suitable for deployment in resource-constrained environments such as hostels, small

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PROJECT AUDIT REPORT

This is to certify that the project work entitled "**SmartVision: A Deep Learning and IoT-Based Framework for Automated Kitchen Hygiene Monitoring**" categorized as an internal project done by "**RAJESH KANNAN L(99220041561), SURARAPU ASHISH S(99220041579), JEYA MITHRA K(99220041618), MEENAKSHI B (99220041666)"** of the Department of Computer Science and Engineering, under the guidance of **Dr.T.Dhiliphan Rajkumar** during the Even semester of the academic year 2025 – 2026 are as per the quality guidelines specified by IQAC.

Quality Grade

Deputy Dean (IQAC)

Administrative Quality Assurance **Dean (IQAC)**